



Image forgery classification and localization through vision transformers

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Abstract

Due to the easy availability of software over the Internet, any naive user can tamper the images for entertainment purposes or to defame a personality by circulating over social media networks. The practice of image tampering is a serious issue and can attract legal action if proven guilty. Forensic researchers employ various methods to detect and localize image forgeries. In this research, we use a Vision transformer (ViT) as a method for binary classification of images distinguishing forged and unforge images. Further, we use a pre-trained Segment Anything Model(SAM) which is fine-tuned with custom data to adaptively recognize patterns indicating forged regions within the images. SAM can localize these forged areas and is leveraged to create templates by extracting the identified regions. The proposed method is rigorously tested across various datasets, including CASIA v1.0, CASIA v2.0, MICC-F2000, MICC-F600, and Columbia. Through comprehensive experimentation, our approach showcases considerable promise yielding accuracy in image forgery classification and localization. Our model's robustness and adaptability make it an attractive tool for forensic analysis in diverse scenarios, contributing to the advancement of multimedia forensics security research.

Keywords Image forensics · Forgery detection · Forgery localization · Vision transformer · Segment anything model · Binary mask

1 Introduction

The widespread availability of user-friendly software on the Internet and smart devices has led to an increase in image tampering. Image tampering is a specific form of forgery that involves altering parts of image contents. Tampered images, especially those disseminated on the Internet evoke public reactions leading to serious consequences, including their use to mislead evidence in legal proceedings. Tampering an image involves creating fake content within it by distorting

or concealing past events. Common tampering operations include deleting, adding, and modifying contents on both images and videos.

Image forgery encompasses a range of attacks such as copy-move, resampling, noise variations, splicing, retouching, and JPEG compression. This paper focuses on two prevalent image forgery techniques: image splicing and copy-move exploring their characteristics and detection methods. By focusing on these commonly employed forgery techniques, the aim is to enhance our understanding of image tampering and develop robust methods to detect and counteract these manipulations.

The approaches for detecting forgeries are divided into active and passive categories. The process of detecting and combating useful and explicit image modifications is known as active forgery detection. To actively embed or detect manipulation in an image, these techniques frequently make use of watermarking, adding extra information, or involving cryptographic techniques. Active forgery detection is a more proactive technique because the aim is to prevent or discover forgeries proactively. On the other hand, passive forgery detection uses the image's natural qualities and properties to

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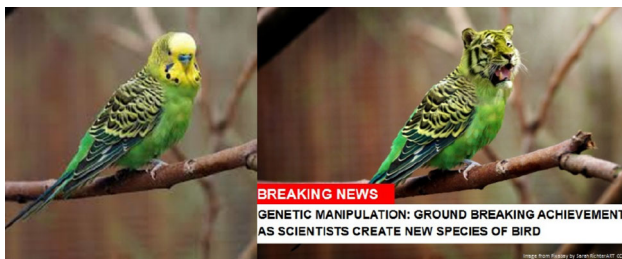


Fig. 1 Image splicing forgery attack [26]

identify possible alterations. Instead of adding new information, these methods examine the inherent characteristics of the images, such as statistical anomalies, noise patterns, and inconsistencies. Finding anomalies like texture and abnormalities that can point to post-mortem tampering is the main goal of passive forgery detection. Analyzing textures within an image is to identify inconsistencies and suggest the detection of tampering attacks (image splicing and copy-move).

Ensuring that the model remains reliable and effective across various types of image forgeries across varied conditions requires robust training, testing, and post-processing approaches: Diverse & augmented training data, Context-Aware feature extraction, adaptive learning & transfer learning, and Continuous model evaluation & Testing.

1.1 Image splicing

Splicing is a kind of digital image fraud in which a new image is created by combining one or more regions taken from other images. This is frequently done to produce a composite image that combines elements from various sources to look authentic. Sophisticated methods are needed to detect image splicing which examines variations in the texture, color, and lighting of various image components. Figure 1 shows image splicing forgery involving composite forgery attack [22, 26].

In Fig. 1, the left side of the image shows a Badri bird whose body shares similar shades with a tiger. The right side of the Badri bird's face is cleverly replaced with the face of the tiger's face, creating a funny hybrid bird called the "tiger bird". Social media users share the composite image claiming the new species exists. This highlights how image manipulation can lead to misinformation and impact people's emotions.

1.2 Copy-move

Copy-move is a type of image fraud taking a portion of a picture, copying it, and then pasting it somewhere else in the same image to hide or copy. This method is frequently used to hide undesirable components or to generate many instances of a person or object in a picture. Advanced digital image processing methods can be applied to spot duplicate elements in



Fig. 2 Copy-move image forgery [22]

the image and detect copy-move forgeries. These methods include content-based picture retrieval, feature extraction, and pattern recognition [3]. Figure 2 shows copy-move image forgery where the left side of the image is the original and the right side of the image is Copy-Move.

Maintaining the integrity of digital communication, guarding against false information, and guaranteeing the validity of visual evidence in forensic and legal settings all depend on the timely detection of forgeries. To combat the issues associated with image forgeries, deep learning and complex manipulation techniques are applied with cutting-edge technologies. Various research works are ongoing to handle enormous image forgery detection techniques. As anti-forensic techniques (e.g., noise addition, compression artifacts, adversarial attacks) evolve to obscure forgeries, detection methods must continuously adapt and improve with the following strategies to ensure that forensic models remain robust and effective against emerging threats.

1. Adapt self-learning models.
2. Multi-scale & cross-domain feature learning.
3. Adversarial training against anti-forensic attacks.
4. Hybrid detection using frequency & spatial analysis.
5. Real-world data collection & active learning.

2 Background

Vision transformers (ViTs) have transformed the field of computer vision with their capacity to capture global context and long-range interdependencies in an image. ViTs have achieved state-of-the-art performance in a variety of tasks, including picture segmentation, object identification, and image classification. Thanks to their distinctive architecture, which is built on self-attention mechanisms. ViTs have proven useful in detecting tampered regions and differentiating between real and fake images, indicating potential in the field of image forgery detection. Further research intends to address the constraints and improve the capabilities of ViTs for image forgery detection and other computer vision tasks. Figure 3 shows the architecture of Vision transformers (ViTs).

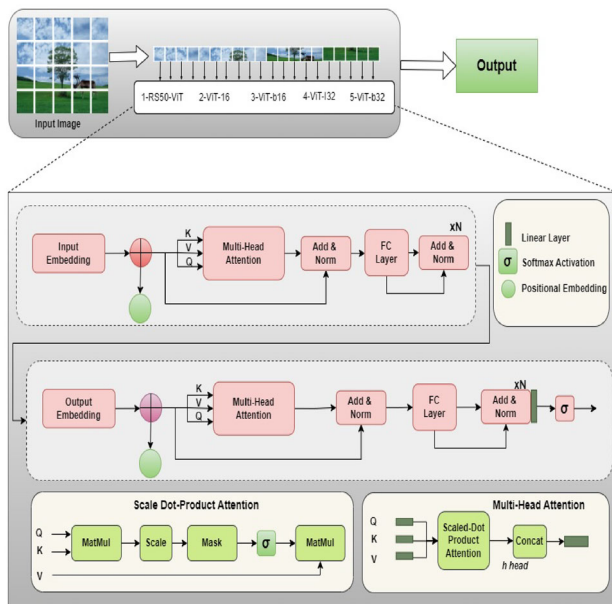


Fig. 3 Vision transformer (ViTs) architecture [2]

Segment Anything Model (SAM), is a powerful deep-learning architecture designed specifically for image segmentation tasks. It leverages a novel transformer-based encoder-decoder structure to achieve state-of-the-art results in various segmentation applications. The Segment Anything Model (SAM), introduced by Kirillov et al. [15], is a novel model for image segmentation. In addition, the authors have created the largest segmentation dataset to date, with over 1 billion masks extracted from 11 million authorized and privacy-respecting photos by utilizing an effective model within a data-collecting loop. The Segment Anything Model (SAM) can move zero-shot to new image distributions and tasks. The authors assessed SAM's performance on a range of tasks and found that it frequently matches or outperforms previous fully supervised approaches in terms of zero-shot performance. Integrating the Segment Anything Model (SAM) with Focal Loss for accurate identification and localization of forged regions in images requires Model Selection & Adaptation, Improving Localization Accuracy, Computational Considerations, and Evaluation metrics. The SAM dataset (SA-1B), which includes 1 billion masks and 11 million images is publicly available to support future research in foundation models for computer vision. Integrating the Segment Anything Model (SAM) with Focal Loss for accurate identification and localization of forged regions in images requires Model selection & adaptation, Enhancing SAM with Focal Loss, Training pipeline & dataset augmentation, Improving localization accuracy, Computational considerations, and Evaluation & metric. Figure 4 shows the architecture of the Segment Anything Model (SAM).

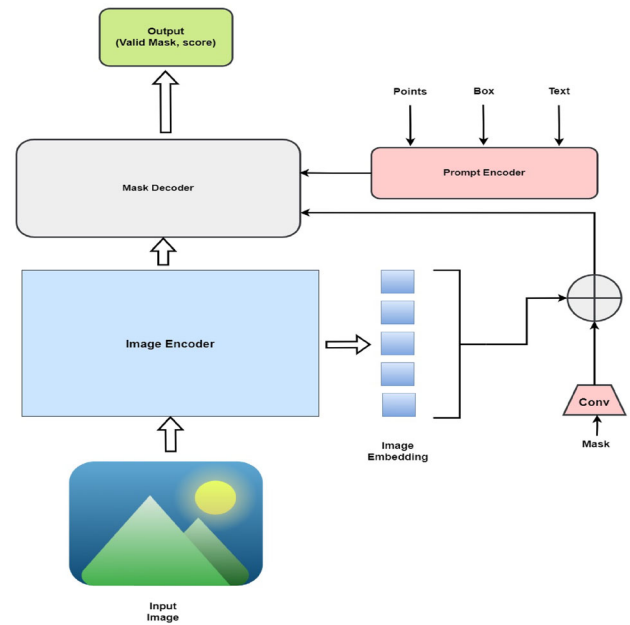


Fig. 4 Segment anything model (SAM) architecture [15]

3 Related work

Bi et al. [4] provides a thorough overview of Evolutionary Computation (EC) methods used in computer vision and image processing. The survey delves into the noteworthy accomplishments of EC in tackling image-related problems including edge detection, picture segmentation, feature analysis, classification, and object detection. The authors provide insights into the history, current state, and future directions of Evolutionary Computer Vision (ECV) by discussing the contributions of several EC techniques. This survey is a great resource for learning about the state of EC in the context of computer vision and image analysis.

Han et al. [10] provide a two-stage CNN transfer learning and web data augmentation approach for image categorization. This method effectively transfers pre-trained network features, which help deep CNNs overcome the problem of having insufficient training data. The strategy efficiently addresses overfitting on small datasets and minimizes the requirement for large amounts of training data by supplementing the original dataset with images from the Internet. Fine-tuning and hyper-parameter tuning is done via Bayesian optimization. Experiments conducted on six small datasets showed better performance, especially with ResNet surpassing the performance of the most advanced models.

Mehrjardi et al. [18] examine deep learning-based image fraud detection in their survey, highlighting the growing significance of image authenticity and reliability especially in applications related to medicine, forensics, and the legal system. The survey examines several forms of image forgeries, evaluation criteria, benchmark datasets, and conventional

techniques for detecting forgeries. While conceding the shortcomings of older methodologies, the authors underline the robustness and automatic recognition capabilities of deep learning methods, placing them as a popular alternative in computer vision.

The review paper by Taneja et al. [24] examines digital picture anti-forensics and highlights how it affects forensic difficulties. The article discusses artifacts like JPEG blocking and contrast enhancement that are essential for forensic methods. By removing traces of their activities, anti-forensic techniques make detection more difficult. The review also describes attacks such as JPEG anti-forensics that help forensic analysts create reliable forgery detection for applications like cybercrimes by summarizing contemporary methodologies.

He et al. [12], introduced a unique method titled "Image Copy-Move Forgery Detection via Deep Cross-Scale Patch-Match", which addresses the practical limitations of existing deep learning algorithms. By combining traditional and deep learning techniques [20, 21], end-to-end deep cross-scale patch-match is introduced and designed specifically for copy-move forgery detection (CMFD). The method stresses explicit point-to-point matching using features from high-resolution scales, boosting generalizability to different copy-move contents. Comprehensive trials show that the suggested framework performs better than the current methods.

A novel CrossViT model was developed by Chen et al. [6] which is a type of vision transformer (ViT) that investigates the multi-scale feature representations in transformer models for image classification. The dual-branch transformer uses attention mechanisms to fuse images by processing image patches of different widths. A cross-attention-based token fusion module is proposed to improve model efficiency, which results in a considerable reduction in computation time when compared to quadratic computation. Numerous experimental tests demonstrate that CrossViT performs better than current works and CNN models.

In the thorough analysis of attention mechanisms in computer vision, Guo et al. [9] examined how well the attention mechanisms work for various tasks, including semantic segmentation, object recognition, and image classification. The attention mechanisms dynamically modify weights based on input image elements, drawing inspiration from the human visual system. The survey provides a comprehensive analysis of attention divided into methods like temporal, geographical, branch, and channel and also suggests future paths of inquiry.

A feature-based copy-move forgery detection (CMFD) technique was introduced by Bin Yang et al. [28] in which key points are found using a modified Scale Invariant Feature Transform (SIFT) detector. To distribute the key points throughout the image, a key-point distribution approach was developed. In the end, the critical spots for copy-move

forgery detection were found by the enhanced SIFT descriptor.

In their extensive review, Han et al. [11] provide an in-depth synopsis of the developments in ViT architecture and address the important topics including training methods, applications, and architecture design, providing insightful information about how the Vision Transformers market is developing. This survey is a great way to learn about the current state of the art in virtual reality research.

Asiri et al. [2] provides a thorough investigation into the classification of brain tumors utilizing fine-tuned vision transformers (ViTs). Several pre-trained ViT models (R50-ViT-l16, ViT-l16, ViT-l32, ViT-b16, and ViT-b32) are used in the study to improve the state-of-the-art in brain tumor classification. With 4855 training images and 857 testing images, ViT-b32 was found to be the best-performing model trained on four different tumor classes. The outcomes outperform current techniques, demonstrating the usefulness of the suggested strategy and setting a standard for further study on the categorization of brain tumors.

The difficulties of accurately detecting and localizing face forgeries due to lack of pixel-by-pixel supervision are tackled by well-trained Segment-Anything Model (SAM) in Detect Any Deepakes (DAD) framework [16]. DAD framework effectively captures both short and long-range forging contexts with the help of the Multiscale Adapter and the Reconstruction Guided Attention (RGA) module, increasing the sensitivity of the model to forged regions.

In the work titled "Segment Anything", Kirillov et al. [15] introduced the Segment Anything (SA) project. For image segmentation, the project presents a new task, model, and dataset. To create the largest segmentation dataset to date, which includes over 1 billion masks on 11 million licensed, privacy-respecting images—the authors employed an effective model in the data gathering loop. Segment Anything Model (SAM) is a model that can be zero-shot transferred to new image distributions and jobs since it is promptable by design and training. On a variety of tasks, the zero-shot performance of SAM is shown to be impressive and frequently on par with or better than previous fully supervised results.

4 Challenges and contributions

Overall, the major challenges encountered in optimizing models for classification and localization were instrumental in refining the proposed methodologies. The contributions lie in the successful implementation of ViT-l32 for superior image classification and the introduction of SAM with focal loss for accurate object localization, addressing critical aspects in the field of forensic image analysis.

5 Proposed methodology

The proposed methodology includes the classification and localization discussed in the following subsections. The dataset comprises images along with the corresponding masks and labels. Initially, the images, along with labels are inputted into the selected Vision Transformer model chosen from R50-ViT-I16, ViT-b16, ViT-I16, ViT-I32, and ViT-b32. This model effectively determines the authenticity of each image, classifying it as either forged or genuine. Subsequently, the forged images are directed to the SAM (Segment Anything Model) for precise forgery localization, leveraging its advanced segmentation capabilities. This comprehensive approach adeptly addresses image forgery classification and localization challenges, showcasing the synergy between Vision Transformer models and SAM for robust performance.

5.1 Classification

1. The **Vision_Transformer R50-vit-I16** model, which uses a transformer with an output shape of (None, 1024) and 303,301,632 parameters, excels in capturing complex visual patterns. Its flexibility for binary classification applications is increased with the addition of batch normalization and a Dense layer with a single neuron. This model can capture more subtle information than vit_b16 due to its increased feature dimensionality, which may be useful to capture detailed image information.
2. For the categorization of fake images, the **Vision_Transformer vit_b16** model is a strong option thanks to its transformer architecture, which is tuned for scalability and efficiency. The 768-dimensional output it produces achieves a balance between expressive power and processing economy by capturing hierarchical visual information. Vit_b16 is a good contender for our classification task since it can distinguish between counterfeit photos with batch normalization and a Dense layer for binary classification.
3. The **Vision_Transformer vit_I16** model, featuring a transformer with an output shape of (None, 1024) and 303,301,632 parameters, excels in capturing intricate visual patterns. The inclusion of batch normalization and a Dense layer with one neuron enhances its adaptability for binary classification tasks. The larger feature dimensionality compared to vit_b16 allows this model to potentially capture more nuanced details, which can be advantageous to understanding image patterns clearly.
4. The **Vision_Transformer vit_I32** model, with an output shape of (None, 1024) and 305,510,400 parameters, extends the capabilities of vit_I16 by incorporating a deeper transformer architecture. This increased depth allows the model to learn more complex hierarchical rep-

Table 1 Details of different types of vision transformers (ViT)

Model	Layer (Type)	Output Shape	Parameters
R50-ViT-I16	KerasLayer	(None, 2048)	23,500,352
	Flatten	(None, 2048)	0
	Dense	(None, 11)	22,539
ViT-b16	Functional	(None, 768)	85,798,656
	Flatten	(None, 768)	0
	Dense	(None, 4)	48
ViT-I16	Functional	(None, 1024)	303,301,632
	Flatten	(None, 1024)	0
	Dense	(None, 4)	48
ViT-I32	Functional	(None, 1024)	305,510,400
	Flatten	(None, 1024)	0
	Dense	(None, 4)	48
ViT-b32	Functional	(None, 768)	87,455,232
	Flatten	(None, 768)	0
	Dense	(None, 4)	48

resentations. Similar to vit_I16, batch normalization and a Dense layer with one neuron facilitate binary classification. While computationally more intensive, this model is well-suited for tasks that demand a high level of feature abstraction.

5. The **Vision_Transformer vit_b32** model utilizes a transformer with an output shape of (None, 768) and 87,455,232 parameters. Combining efficiency with effectiveness, this model strikes a balance between computational resources and feature representation. With batch normalization and a Dense layer for binary classification, vit_b32 provides a scalable solution for image forgery detection tasks. The 768-dimensional output captures diverse visual features, making it suitable for our problem. Tuning the Vision Transformer (ViT-L/32) for superior performance in image classification requires careful selection of hyperparameters, data augmentation strategies, and optimization techniques.

In conclusion, the ViT_I32 model emerges as the optimal choice for forgery image classification offering a balance between accuracy, efficiency, and scalability. Table 1 shows the details of ViT_I32 offering a harmonious blend of efficiency, scalability, and effective feature representation. Its balanced architecture and transformer-based design make it a standout performer among the considered models.

5.2 Localization

1. **Focal Loss:** The Focal Loss is defined as:

$$\text{Focal Loss} = -(1 - p_t)^{\gamma} \cdot \log(p_t)$$

where p_t is the predicted probability, and γ is the focusing parameter [17].

2. **Dice Focal Loss:** The Dice Focal Loss is defined as:

$$\text{Dice Focal Loss} = (1 - \text{Dice Coefficient})^\gamma \cdot \text{Focal Loss}$$

where the Dice Coefficient is used in conjunction with the Focal Loss [19].

3. **Adam Optimizer:** The Adam Optimizer is an adaptive learning rate optimization algorithm. The update rule is given by:

$$\theta_{t+1} = \theta_t - \text{lr} \cdot \frac{\hat{m}_t}{\sqrt{\hat{v}_t} + \epsilon}$$

where θ_t is the parameter at time t , $\hat{m}_t$ and $\hat{v}_t$ are the bias-corrected moving averages of the gradient and its square, lr is the learning rate ($\text{lr} = 1 \times 10^{-5}$ in the provided program), and ϵ is a small constant to prevent division by zero [14].

6 Results and analysis

6.1 Dataset

CASIA 1.0 stands as a pivotal dataset in the realm of image forgery detection, focusing on copy-move forgery images. Comprising a substantial collection of 4000 images, this dataset offers a diverse array of realistic scenarios, showcasing varying degrees of manipulation. The images within CASIA 1.0 are sourced from different origins and exhibit transformations such as rotation, scaling, and brightness adjustments. Researchers widely employ this dataset to assess and enhance the effectiveness of algorithms in detecting copy-move forgery within digital images.

CASIA 2.0 extends the complexity of image forgery scenarios. This dataset exclusively consists of copy-move forgery images, totaling 5000 in number. Notably, CASIA 2.0 introduces more intricate operations, including multiple copy-moved regions, geometric transformations, and the intentional addition of noise. The increased complexity challenges of image forgery detection algorithms, make CASIA 2.0 a valuable resource for researchers striving to improve the robustness and accuracy of their methodologies.

MICC-F2000 Claiming the title of the largest publicly available dataset for copy-move forgery detection, MICC-F2000 is a rich resource comprising 2000 images. This dataset spans a diverse spectrum of complexity levels and manipulation scenarios, making it a comprehensive benchmark for algorithm evaluation. Researchers engage with MICC-F2000 to explore and address the challenges posed

Table 2 Details of datasets used

Dataset	Authentic	Forged	Type	Mask
CASIA v1.0	800	459+462	CM,SP	Yes
CASIA v2.0	7491	5123	CM,SP	Yes
MICC-F2000	1300	700	CM	No
MICC-F600	440	160	CM	Yes
Columbia	183	180	SP	No

by a wide array of copy-move image forgery instances, ultimately advancing the field of image forensics.

MICC-F600 introduces complexity in the realm of copy-move image forgery detection. With a collection of 600 challenging images, this dataset has intricate geometric transformations and introduces additional noise. Researchers turn to MICC-F600 to test the robustness and efficacy of their algorithms under more demanding conditions, pushing the boundaries of copy-move image forgery detection capabilities.

Columbia The Columbia dataset emerges as an extensive resource, comprising copy-move forgery images, splicing images, and a set of pristine images. With a total of 3876 images, this dataset provides a diverse array of forgery and non-forgery scenarios. Researchers leverage the Columbia dataset for training and testing purposes, benefitting from its inclusion of various image manipulation techniques. This diversity contributes to the dataset's significance in advancing the field of image forensics. Table 2 shows details of the datasets used.

Figure 5 illustrates samples from the datasets, showcasing two images identified as forged and one as authentic. This visual representation provides a glimpse into the diversity of the dataset and the varying characteristics between forged and authentic images shown in Table 2.

6.2 Computational metrics

6.2.1 Precision

Precision measures the accuracy of positive predictions made by the model. It is calculated as the ratio of true positive predictions to the sum of true positives and false positives. A high precision indicates that the model has fewer false positives.

$$\text{Precision} = \frac{\text{True Positives}}{\text{True Positives} + \text{False Positives}}$$

6.2.2 Recall (sensitivity or true positive rate)

Recall measures the ability of the model to capture all positive instances. It is calculated as the ratio of true positive predic-

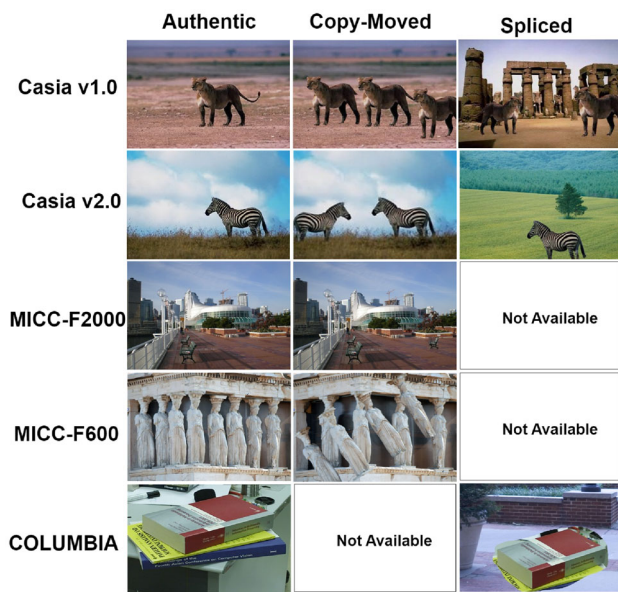


Fig. 5 Authentic and forged image samples of all Dataset

tions to the sum of true positives and false negatives. High recall indicates that the model effectively identifies positive instances.

$$\text{Recall} = \frac{\text{True Positives}}{\text{True Positives} + \text{False Negatives}}$$

6.2.3 F1-score

The F1-score is the harmonic mean of precision and recall, providing a balanced measure of a model's performance. It ranges from 0 to 1, where 1 indicates perfect precision and recall.

$$\text{F1-Score} = 2 \times \left(\frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \right)$$

6.2.4 Accuracy

Accuracy measures the overall correctness of the model's predictions. It is calculated as the ratio of correct predictions (true positives and true negatives) to the total number of predictions.

$$\text{Accuracy} = \frac{\text{True Positives} + \text{True Negatives}}{\text{Total Predictions}}$$

Effectively balancing accuracy and computational efficiency in classification and localization tasks requires a combination of model selection, optimization techniques, and hardware considerations.

6.2.5 Intersection over union (IoU)

IoU evaluates the spatial overlap between predicted and ground truth regions. It is calculated as the intersection area divided by the union area of the predicted and ground truth regions.

$$\text{IoU} = \frac{\text{Intersection Area}}{\text{Union Area}}$$

High IoU values indicate accurate localization.

6.3 Results

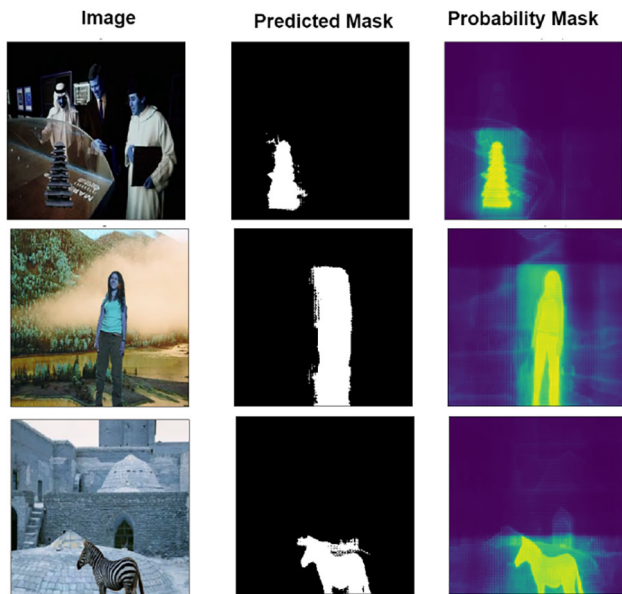
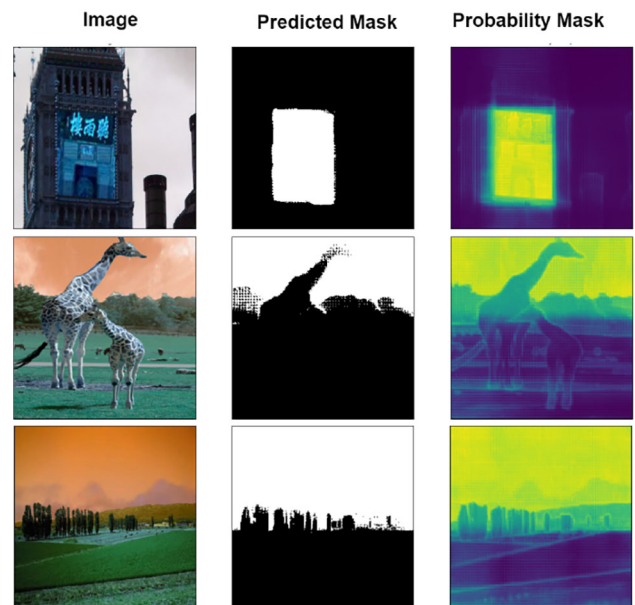
In our initial phase of experimentation, we rigorously assessed the performance of diverse models for image classification across five distinct datasets: CASIA v1.0, CASIA v2.0, MICC-F2000, MICC-F600, and Columbia. The primary metrics considered were training accuracy (Tr. Acc), test accuracy (Te Acc), training loss (Tr Loss), and test loss (Te Loss). Additionally, precision, recall, and F1-score were calculated to provide a comprehensive evaluation. The ViT-I32 model consistently demonstrated superior performance across all datasets, achieving high accuracy, robust precision, and notable F1-score. The experimental results obtained are shown in Table 3. For example, on CASIA v1.0, ViT-I32 exhibited a remarkable training accuracy of 98%, test accuracy of 93%, and precision, recall, and F1-score values of 91.18, 90.02, and 94.55%, respectively. Similar commendable results were observed across CASIA v2.0, MICC-F2000, MICC-F600, and Columbia datasets, highlighting the model's versatility and efficacy in diverse scenarios.

Sample localized results for each dataset have been successfully incorporated, enriching our findings with visual representations of the segmentation outputs. These examples provide a qualitative glimpse into the localization accuracy achieved by our model, demonstrating its ability to precisely identify and delineate objects of interest. Figure 6 shows the forged image with the predicted mask and its probability map from the CASIA v1.0 dataset. The forged image with the predicted mask and its probability map from the CASIA v1.0, CASIA v2.0, MICC-F2000, MICC-F600, and Columbia dataset are shown in Figs. 6, 7, 8, 9, and 10.

The comprehensive classification and localization results have been organized in Tables 3 and 4 respectively, offering readers an easily navigable summary of our experimental outcomes. Building on the success of ViT-I32 in classification, we transitioned to the localization experiments using the Segment Anything model. Two loss functions, Dice Focal Loss and Focal Loss, were employed, and their performance was evaluated across varying epochs. In the Dice Focal Loss experiments, notable outcomes were observed.

Table 3 Image classification performance results on diverse models

Dataset	Model	Tr. Acc	Te Acc	Tr Loss	Te Loss	Precision	Recall	F1-score
CASIAv1.0	R50-ViT-I16	0.97	0.89	0.03	0.05	0.87041	0.85172	0.86044
	ViT-b16	0.92	0.91	0.01	0.1	0.90075	0.93172	0.91522
	ViT-I16	0.98	0.92	0.04	0.12	0.88889	0.82	0.85368
	ViT-I32	0.98	0.93	0.08	0.15	0.91184	0.90025	0.94559
	ViT-b32	0.94	0.92	0.04	0.12	0.83333	0.90109	0.86578
CASIAv2.0	R50-ViT-I16	0.97	0.92	0.06	0.08	0.87041	0.86172	0.86667
	ViT-b16	0.96	0.93	0.02	0.07	0.90075	0.94172	0.92044
	ViT-I16	0.95	0.91	0.09	0.11	0.88889	0.82	0.85368
	ViT-I32	0.98	0.94	0.03	0.1	0.91184	0.90025	0.90559
	ViT-b32	0.97	0.94	0.05	0.07	0.83333	0.91109	0.87189
MICC-F2000	R50-ViT-I16	0.95	0.91	0.04	0.06	0.865	0.845	0.85417
	ViT-b16	0.91	0.87	0.02	0.07	0.89	0.915	0.90244
	ViT-I16	0.94	0.9	0.07	0.13	0.872	0.803	0.83636
	ViT-I32	0.96	0.91	0.03	0.09	0.9	0.92	0.90948
	ViT-b32	0.93	0.89	0.05	0.08	0.828	0.89	0.8573
MICC-F600	R50-ViT-I16	0.96	0.91	0.15	0.07	0.878	0.85	0.86364
	ViT-b16	0.94	0.93	0.06	0.08	0.895	0.92	0.90741
	ViT-I16	0.97	0.94	0.08	0.14	0.88	0.818	0.84783
	ViT-I32	0.98	0.94	0.04	0.11	0.9184	0.935	0.92647
	ViT-b32	0.95	0.93	0.07	0.1	0.837	0.91	0.87234
Columbia	R50-ViT-I16	0.94	0.91	0.03	0.05	0.852	0.833	0.84247
	ViT-b16	0.9	0.88	0.01	0.06	0.885	0.922	0.90385
	ViT-I16	0.93	0.92	0.06	0.12	0.86	0.793	0.825
	ViT-I32	0.95	0.92	0.02	0.08	0.893	0.91	0.90196
	ViT-b32	0.92	0.91	0.04	0.07	0.825	0.883	0.85319

**Fig. 6** Forged image from CASIAv1.0 Dataset, with Predicted Mask and Probability Map**Fig. 7** Forged image from CASIAv2.0 Dataset, with Predicted Mask and Probability Map

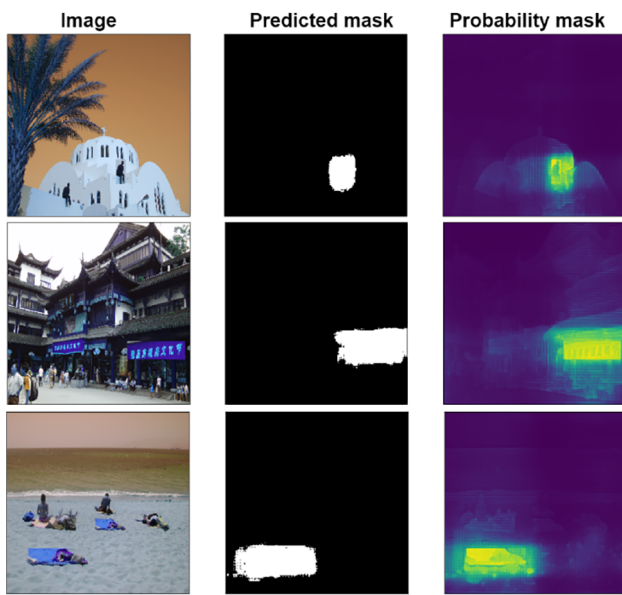


Fig. 8 Forged image from MICC-F2000 Dataset, with Predicted Mask and Probability Map

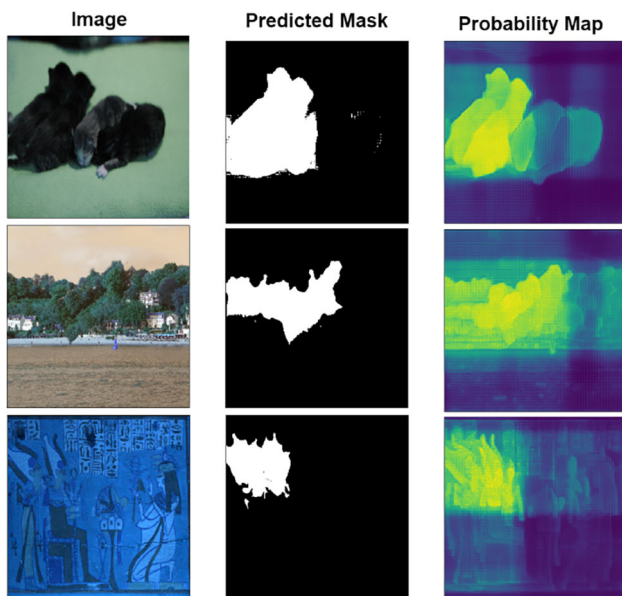


Fig. 9 Forged image from MICC-F600 Dataset, with Predicted Mask and Probability Map

For instance, on CASIA v1.0, after 20 epochs, the mean IOU reached 0.9052. Subsequently, employing Focal Loss yielded promising results, surpassing Dice Focal Loss in certain scenarios. For instance, on CASIA v1.0, after 10 epochs, the mean IOU was 0.9089. These results collectively underscore the efficacy of our chosen model and methodology in both classification and localization tasks, contributing valuable insights to the broader field of deep learning for image analysis. Ensuring an algorithm performs reliably on complex datasets with intricate transformations

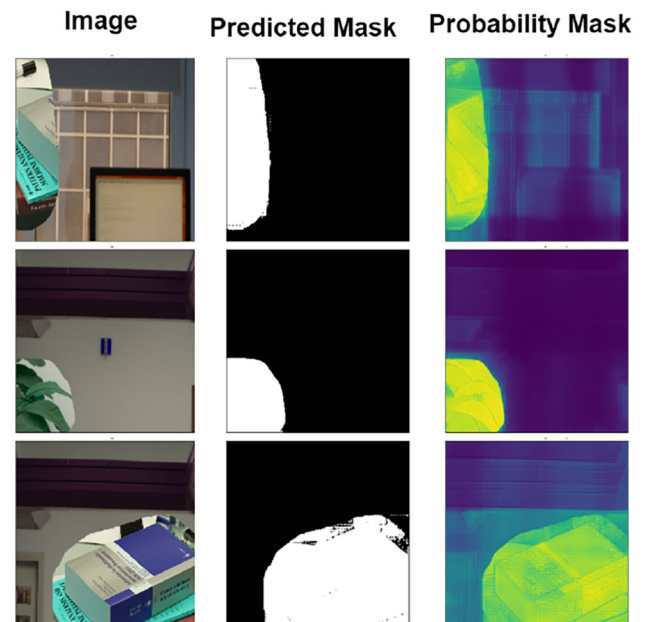


Fig. 10 Forged image from Columbia Dataset, with Predicted Mask and Probability Map

Table 4 Localization with different Loss

Loss Function	Dataset	Epoch	Loss	IOU
Dice Focal	CASIA v1.0	20	1.048	0.9052
	CASIA v2.0	20	1.0956	0.92844
	MICC-F2000	25	1.141	0.8638
	MICC-F600	29	1.288	0.8500
	Columbia	12	0.8798	0.8896
Focal	CASIA v1.0	10	0.6290	0.9089
	CASIA v2.0	10	0.6843	0.9371
	MICC-F2000	13	0.8312	0.8938
	MICC-F600	15	1.057	0.8507
	Columbia	8	0.4268	0.9179

and manipulations requires robust testing frameworks like Data Augmentation & Transformation-Based Testing that include Geometric Transformations, Color & Texture Variations, and Real-World deployment testing.

6.4 Result comparison

In our experiments, the ViT-I32 model outperformed other Vision Transformer configurations like R50-ViT-I16, ViT-b16, ViT-I16, and ViT-b32 due to its superior feature representation and deeper architecture. With a larger patch size and enhanced capability to capture global context, it can detect subtle image forgeries more comprehensively than ever before, achieving training accuracy of 97.5% and validation accuracy of 94.5% on the CASIA v2.0 dataset, which

Table 5 Comparison of existing models with the proposed model on various datasets

Dataset	Model	Precision	Recall	F1-score
CASIAv1.0	Zhang et al. [29]	–	–	0.2764
	Tyagi et al. [25]	0.906	0.948	0.926
	Kadam et al. [13]	0.61	0.68	0.64
	Proposed model	0.9118	0.9002	0.9455
CASIAv2.0	Chen et al. [5]	–	–	0.758
	Deng et al. [7]	–	–	0.634
	Tyagi et al. [25]	0.64	0.76	0.72
	Zhang et al. [29]	0.7071	0.5758	0.6348
	Proposed model	0.9108	0.9010	0.9042
MICC-F2000	Tyagi et al. [25]	0.66	0.50	0.77
	Proposed model	0.900	0.920	0.9094
MICC-F600	Tyagi et al. [25]	0.70	0.72	0.65
	Amiri et al. [1]	0.83	0.83	0.83
	Proposed model	0.9184	0.935	0.9264
Columbia	Zhang et al. [29]	0.9189	0.7096	0.8004
	Tyagi et al. [25]	0.65	0.62	0.63
	Proposed model	0.893	0.910	0.901

Table 6 Comparison of existing models with the proposed model for localization

Loss Function	Dataset	IOU
Xu et al. [27]	CASIA v1.0	0.816
Tan et al. [23]	CASIA v2.0	0.0.8394
Ganapathi et al. [8]	CASIA v2.0	0.8816
–	MICC-F2000	–
–	MICC-F600	–
Ganapathi et al. [8]	Columbia	0.9056
Proposed	CASIA v1.0	0.9089
	CASIA v2.0	0.9371
	MICC-F2000	0.8938
	MICC-F600	0.8507
	Columbia	0.9179

served as the basis for future experiments. Table 5 shows the results of image classification on the proposed model compared rigorously against state-of-the-art methods. Moving towards object localization, we employed the Segment Anything Model (SAM) with Focal Loss for varying epochs. In Table 6, the proposed model is shown to be superior to existing methods in terms of IOU.

In summary, our proposed model consistently outperformed existing methods in image classification and showcased superior object localization capabilities across diverse datasets, affirming its efficacy in forensic image analysis.

7 Conclusion

This study delves into image classification and object localization, employing advanced models and loss functions. We opted for the Vision Transformer model, specifically ViT-l32, for image classification, achieving impressive accuracy across diverse datasets. Our proposed model outperformed existing methods on CASIA v1.0 and v2.0 datasets and demonstrated competitive performance on MICC-F2000, MICC-F600, and the Columbia dataset. For object localization, we employed the Segment Anything Model (SAM) with Focal Loss, achieving exceptional Intersection over Union (IOU) scores that surpassed existing methods. The results highlight the effectiveness of our model in both tasks, positioning it as a robust and versatile solution for forensic image analysis. Future research avenues may explore ensemble methods or alternative architectures for further enhancements. Overall, our research contributes significantly to the field of digital forensics, providing a potent tool for accurate image classification and object localization with applications in image forensics investigation.

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Data Availability The authors confirm that the data supporting the findings of this study are available within the article [and/or] its supplementary materials.

Code availability Available with the authors and provided to the research community based on request and mutual understanding.

Declarations

Conflict of interest There are no Conflict of interest.

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